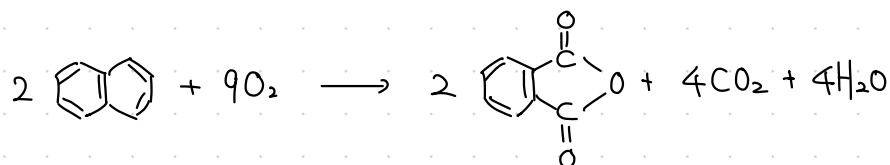
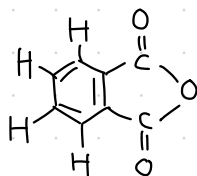
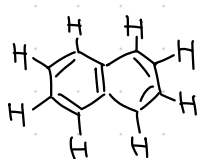
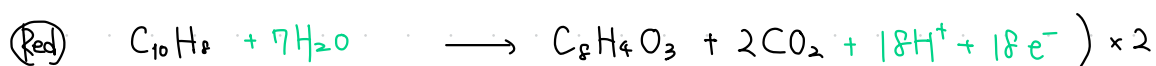
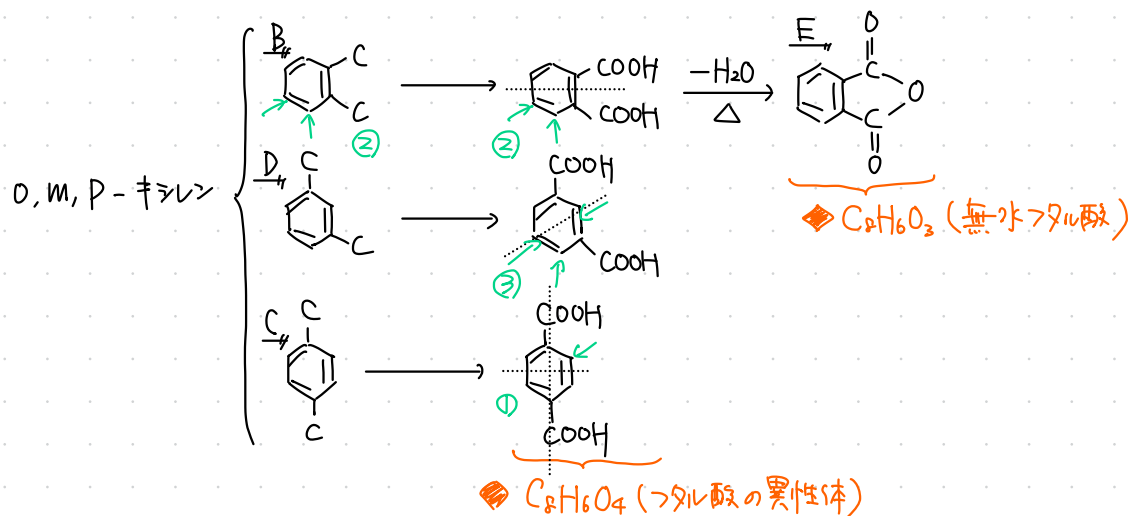
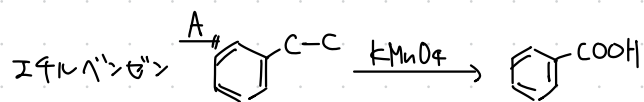


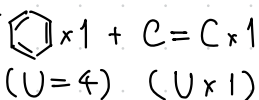
4-① C_8H_{10} $U = 4$ $\text{C}_6\text{H}_5 \times 1$



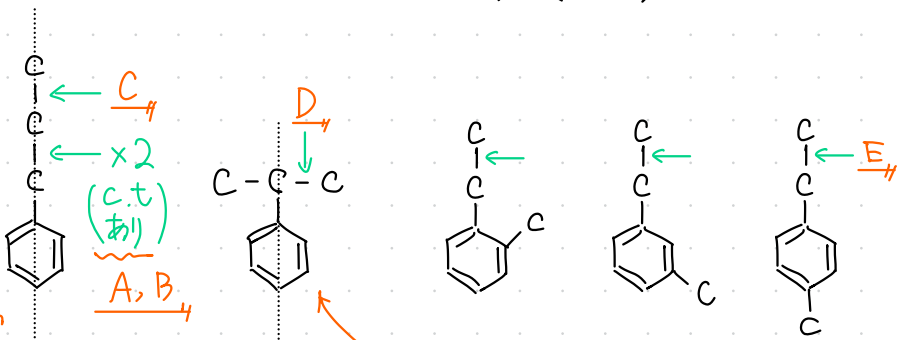
4-2

C_9H_{10}

$U = 5$

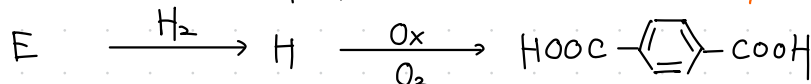
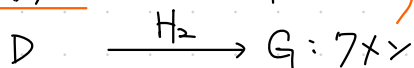


内2.



< 1置換体 >

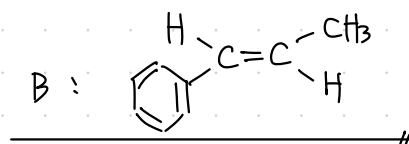
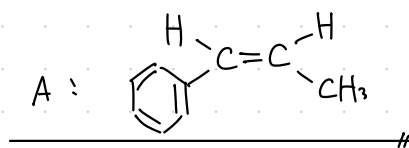
< 2置換体 >



A, B : シス・トランス異性体

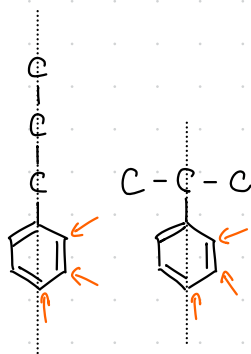
融点 $A < B$

シス トランス



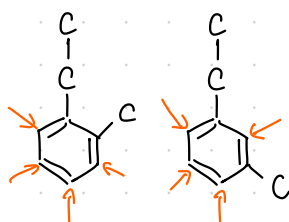
内1. 8

内3. 22



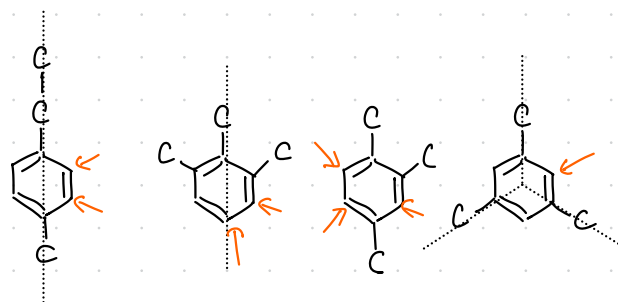
< 1置換体 >

Cl置換: 6



< 2置換体 >

Cl置換: 10



< 3置換体 >

Cl置換: 6

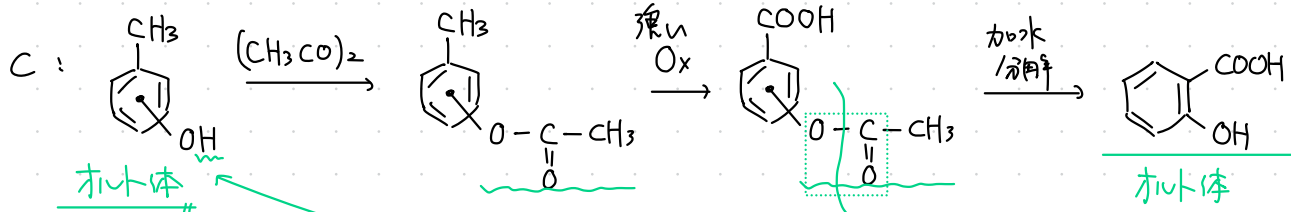
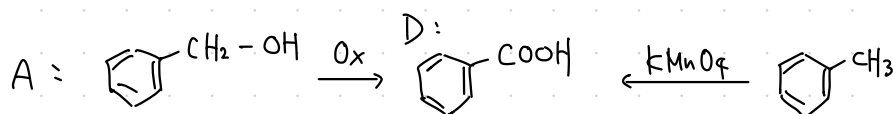
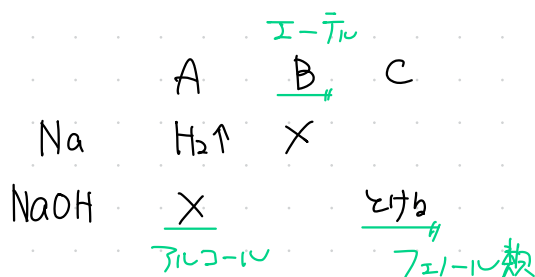
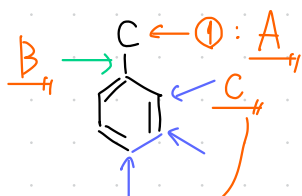
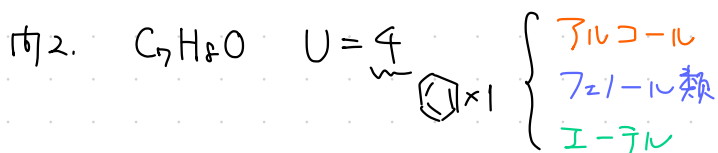
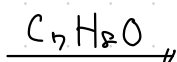
4-③

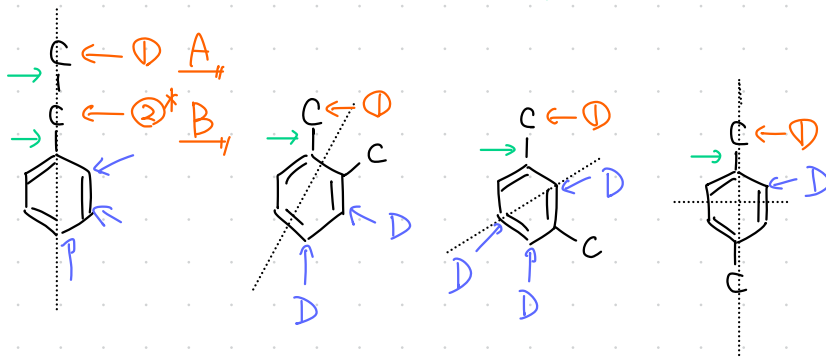
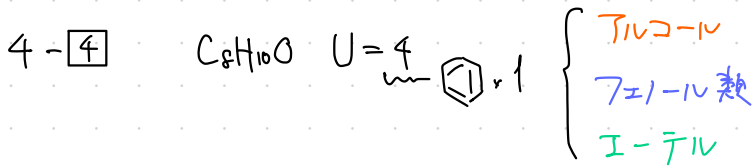


条件より $l : \frac{m}{2} = 7 : 4 \quad \therefore (l, m) = (7, 8)$

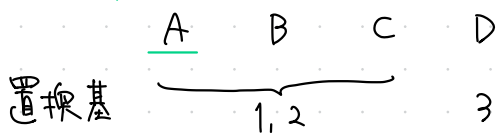
$\therefore (C_7H_8O_n)_k = (92 + 16n)k = 108$

$\therefore (n, k) = (1, 1)$

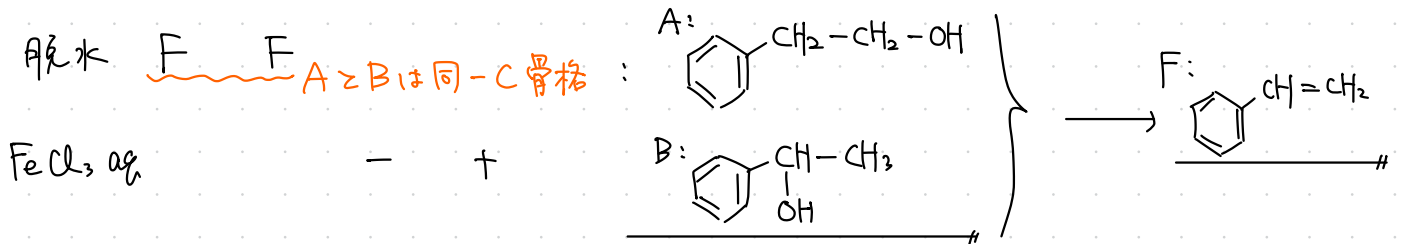




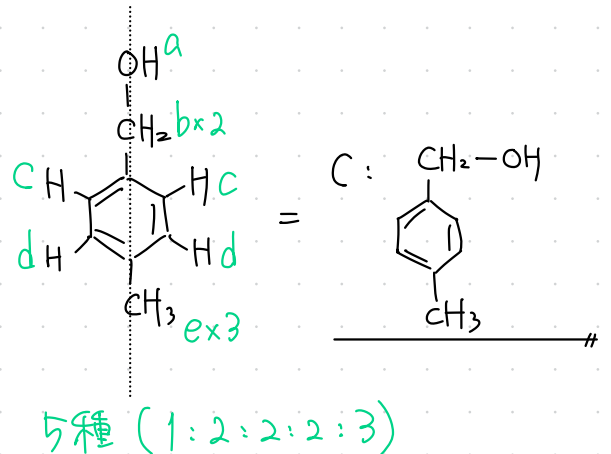
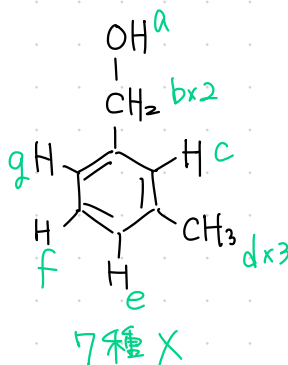
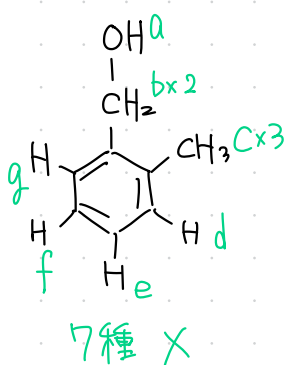
第1級
アルコール



おだやみ
Ox $\begin{array}{c} E \\ \hline \text{アルコール} \end{array}$



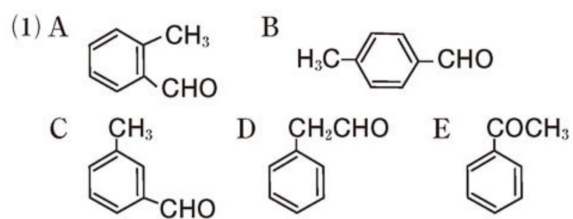
C : 1 or 2 置換体のアルコール



内2

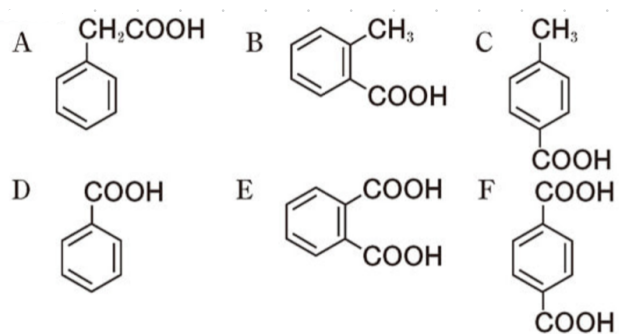
D: フェニル類 (3 置換体), $\underline{6}$

4 - 5

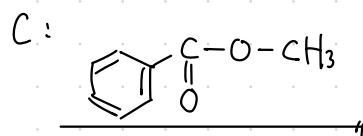
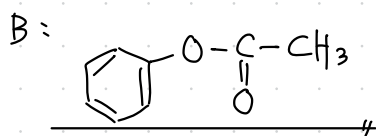
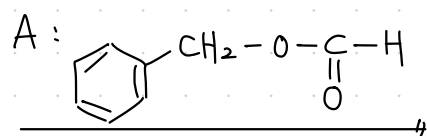
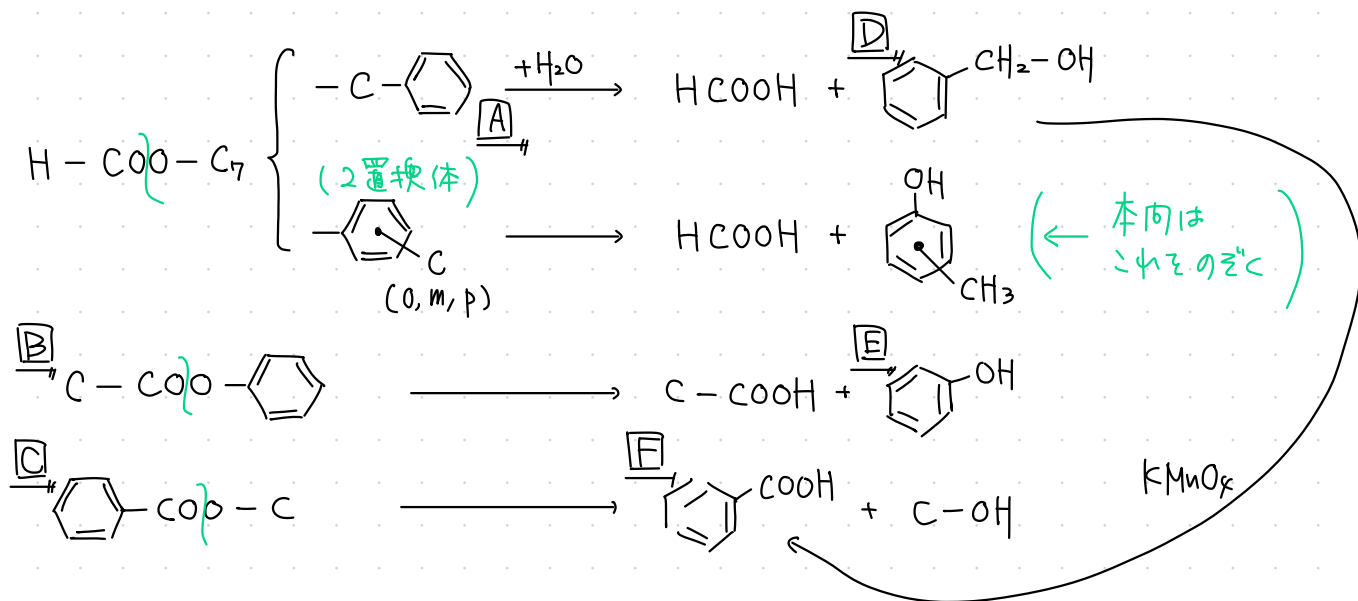
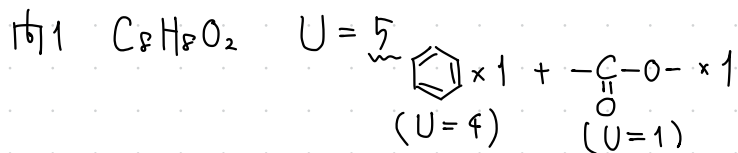


- (2) Gは*p*-置換体であるため、*o*-, *m*-置換体よりも分子の形が対称的である。したがって、結晶化しやすいため、融点は最も高くなる。

4 - 6

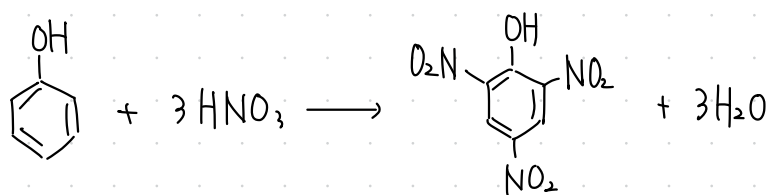


4-7



162. E

163



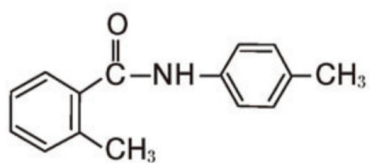
ヒョリニニ

2, 4, 6 - トニトロフェノール

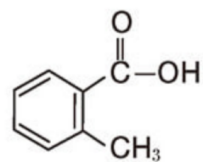
(1) $\text{C}_{15}\text{H}_{15}\text{NO}$

(2)

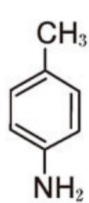
A



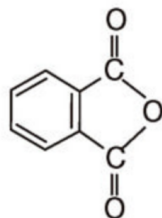
B



C

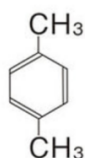


D

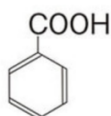


4 - 9

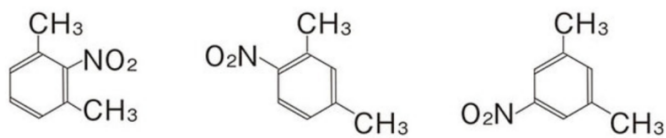
問1 化合物 D :



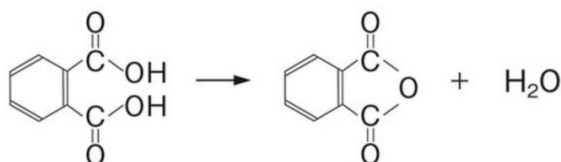
問2 化合物 E :



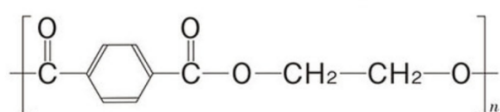
問3



問4



問5

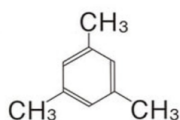


4 - 10

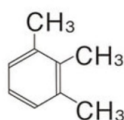
問1 (ア) エチレングリコール(1,2 エタンジオール) (イ) 酸素

(ウ) ヨウ素

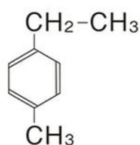
問2 A



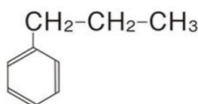
B



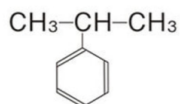
C



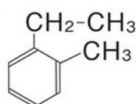
D



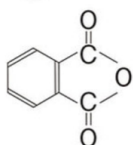
E



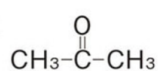
F



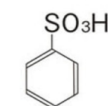
K



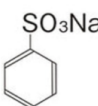
M



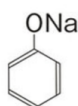
問3 N



O



P

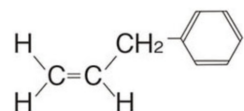


4 - 11

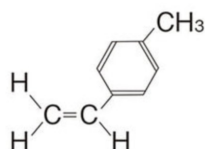
問1 2個

問2 クメン

問3A



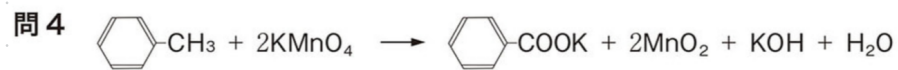
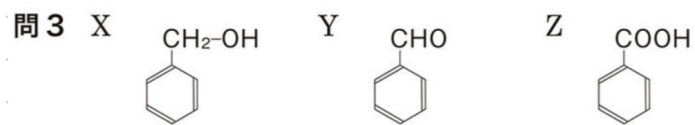
E



4-12

問1 気体…水素 官能基…ヒドロキシ基

問2 試薬… FeCl_3 色…紫

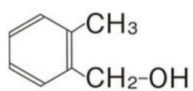


4-13

4-14

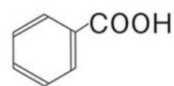
(1) 5

(2)

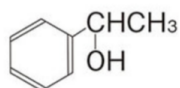


(3) CHI_3

(4)

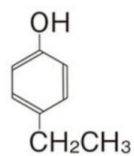


(5)



(6) 9

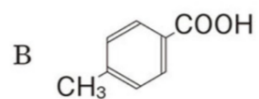
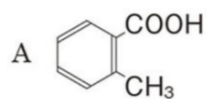
(7)



(8) 5

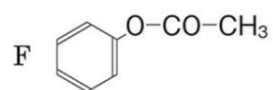
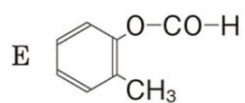
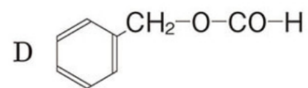
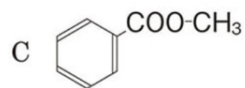
(9) 1 : 2 : 2 : 2 : 3

問 1 (1)



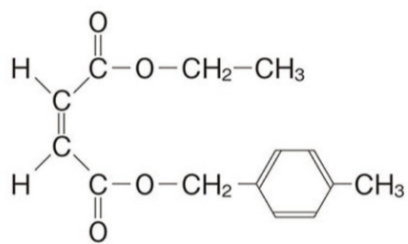
(2) ナフタレン, *o*-キシレン

問 2 (1)

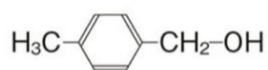


問 1 $C_{14}H_{16}O_4$

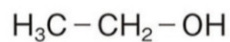
問 2 A



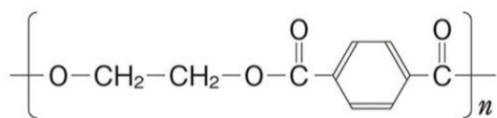
C



D



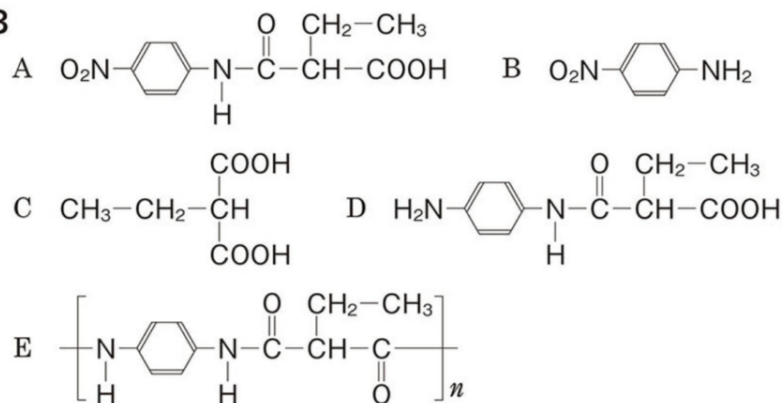
H



問 1 C 原子…52.4 % H 原子…4.8 % N 原子…11.1 %

問 2 $\text{RCOOH} + \text{NaHCO}_3 \longrightarrow \text{RCOONa} + \text{CO}_2 + \text{H}_2\text{O}$

問 3



問 4 塩酸を用いることで化合物 A のアミド結合が加水分解されてしまうため。

